Ofqual QAN	**600/0406/X**
Quartz ID	**1156 / 1157**
Qualification	**Level 4 Diploma in Computing / Level 4 Diploma in Computing (with Business Management)**

Unit ID	**2223**
Assessment	**Computer Systems 2.0**
Assessment Type	**Global Examination**
Date	**Wednesday 4th June 2025**
Time	**2.00pm – 5.00pm**

Marking Scheme

This marking scheme has been prepared as a **guide only** to markers. This is not a set of model answers, or the exclusive answers to the questions, and there will frequently be alternative responses which will provide a valid answer. Markers are advised that, unless a question specifies that an answer be provided in a particular form, then an answer that is correct (factually or in practical terms) **must** be given the available marks.

If there is doubt as to the correctness of an answer, the relevant NCC Education materials should be the first authority.

Throughout the marking, please credit any valid alternative point.

Answer ALL questions
Do NOT copy the questions to your answer booklet

Marks

Question 1

For each answer, only provide the corresponding numeral (e.g., i, ii, iii, or iv).

a) Which component of the computer system is primarily responsible for temporarily holding data **and** instructions for quick access during processing? **1**

i) Cache
ii) Memory Unit
iii) Registers
iv) Random Access Memory

Mark Scheme:
i) Cache (1 mark)

b) In data communications, which of the following is **NOT** considered a goal? **1**

i) Memory capacity
ii) Cost-effectiveness
iii) Word length
iv) Pipelining

Mark Scheme:
ii) Cost-effectiveness (1 mark)

c) What does SMTP stand for in email communication? **1**

i) Secure Mail Transfer Protocol
ii) Secure Messaging Transfer Protocol
iii) Simple Mail Transfer Protocol
iv) Simple Messaging Transfer Protocol

Mark Scheme:
iii) Simple Mail Transfer Protocol (1 mark)

d) Which of the following best describes a fundamental characteristic of quantum computing? **1**

i) Quantum computers use classical bits (1s and 0s) to process information.
ii) Quantum computers rely on quantum bits (qubits) that can exist in multiple states simultaneously.
iii) Quantum computers are slower than traditional computers.
iv) Quantum computers are limited to solving only mathematical problems.

Mark Scheme:
ii) Quantum computers rely on quantum bits (qubits) that can exist in multiple states simultaneously. (1 mark)

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e) What is the primary benefit of CPU pipelining in computer architecture? **1**

i) It increases the clock speed of the CPU.

ii) It minimises the power consumption of the CPU.

iii) It allows multiple instructions to be processed simultaneously in different stages.

iv) It reduces the size of the CPU.

Mark Scheme:
iii) It allows multiple instructions to be processed simultaneously in different stages. (1 mark)

f) Which of the following logic gates produces an output of ONE (1) only when both inputs are different? **1**

i) AND Gate

ii) OR Gate

iii) XOR Gate

iv) NOT Gate

Mark Scheme:
iii) XOR Gate (1 mark)

g) What is the decimal equivalent of the binary number 1011? **1**

i) 8

ii) 9

iii) 10

iv) 11

Mark Scheme:
iv) 11 (1 mark)

h) Which type of CPU addressing mode uses a register's value as a pointer to the memory location of data? **1**

i) Indirect Addressing

ii) Immediate Addressing

iii) Direct Addressing

iv) Indexed Addressing

Mark Scheme:
i) Indirect Addressing (1 mark)

i) Which device is used to connect multiple computers within a local area network (LAN) and manage data traffic between them? **1**

i) Hub

ii) Switch

iii) Modem

iv) Router

Mark Scheme:
ii) Switch (1 mark)

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j) Which system software is responsible for translating high-level programming language into machine code? **1**

i) SQL Database

ii) Operating System

iii) Interpreter

iv) Compiler

Mark Scheme:
iv) Compiler (1 mark)

Total 10 Marks

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Question 2

a) Explain in detail the distinction between hardware and software in a computer system. Use examples to illustrate your answer. **5**

Mark scheme:

- ***Defining hardware as the physical components of a computer system (e.g., CPU, RAM, motherboard, etc.). (1 mark)***
- ***Explanation that hardware is tangible and can be touched. (1 mark)***
- ***Defining software as the programs and instructions that control the hardware. (1 mark)***
- ***Explaining that software is intangible and cannot be physically touched / virtual. (1 mark)***
- ***Explaining that software provides the instructions for the hardware to perform tasks. (1 mark)***

b) Discuss the function of the control unit in relation to the Central Processing Unit. **5**

Mark scheme:

- ***Stating that the Control Unit (CU) is a part of the CPU. (1 mark)***
- ***Explaining that the control unit manages and coordinates the execution of instructions. (1 mark)***
- ***Describing that the CU decodes instructions and controls data flow between the CPU and other components. (1 mark)***
- ***Explaining that the control unit manages the timing of operations through the system clock. (1 mark)***
- ***Explaining how the control unit communicates with other parts of the CPU (e.g., through buses). (1 mark)***

Total 10 Marks

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Question 3

Operating Systems are the 'backbone' of modern computing, facilitating the interaction between the hardware and software components of a computer system.

a) Discuss the following THREE (3) main functions of an operating system with a description of their roles and how each interacts with hardware and software applications. **9**

- File System Management
- Device Management
- Security and Access Control

Mark scheme:

File System Management (Up to 3 marks)

- ***Explaining the OS's role in managing files or directories E.g., storage organization, retrieval, or indexing. (1 mark)***
- ***Describing specific mechanisms for file permissions or protection E.g., controlling user access to files). (1 mark)***
- ***Discussing its interaction with hardware. E.g., HDDs, SSDs, or file allocation OR software (e.g., file sharing between applications).(1 mark)***

Device Management (Up to 3 marks)

- ***Explaining how the OS uses device drivers to communicate with hardware.(1 mark)***
- ***Discussing how input/output (I/O) devices and peripherals are managed by the OS. (1 mark)***
- ***Providing examples of interaction with hardware (e.g., printers, graphics cards) OR software (e.g., enabling device access to applications via APIs). (1 mark)***

Security and Access Control (Up to 3 marks)

- ***Explaining how the OS enforces user authentication (e.g., through passwords, biometrics, or access credentials) or file-level permissions. (1 mark)***
- ***Discussing how these mechanisms ensure controlled access to system resources. (1 mark)***
- ***Providing examples of interaction with hardware (e.g., encryption using hardware modules, secure boot) OR software (e.g., protecting application data or processes). (1 mark)***

b) State ONE (1) additional *main* function of the OS. **1**

Mark scheme:

1 mark for any reasonable answer, not duplicated from q3a:

- ***Memory Management (1 mark)***
- ***Network Management (1 mark)***
- ***User Interface Management (1 mark)***
- ***Error Detection and Handling (1 mark)***

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- ***System Performance Monitoring (1 mark)***
- ***Virtualization Support (1 mark)***

Total 10 Marks

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Question 4

Computer programming is ONE (1) of the core elements of computing and understanding how instructions are processed and executed by a computer is fundamental.

a) Explain the concept of addressing **and** describe the differences between direct, immediate and indirect addressing. Provide at least ONE (1) example of how these modes differ. **5**

Mark Scheme:

<u>Concept of addressing (up to 1 mark):</u>

Addressing modes define how the operands of an instruction are accessed (i.e., how the data for an operation is located). (1 mark)

<u>Descriptions of 3 modes of addressing, and at least one associated example (up to 4 marks):</u>

Immediate Addressing: The operand is provided as part of the instruction itself. (1 mark)

Direct Addressing: The operand's memory address is directly specified. (1 mark)

Indirect Addressing: The memory address of the operand is stored in a register, and the instruction points to that register. (1 mark)

An example showing how these modes differ (e.g., "Immediate gives data directly, Direct accesses memory once, Indirect accesses memory twice"). (1 mark)

b) Explain the key differences between RISC (Reduced Instruction Set Computing) and CISC (Complex Instruction Set Computing) architectures. Highlight ONE (1) distinct advantage and ONE (1) distinct disadvantage for each architecture by focusing on the design principles or operational impact. **4**

Mark scheme:

Explanation of RISC key design principle (up to 1 mark).
Explanation of CISC key design principle (up to 1 mark).
Disadvantage for RISC (up to 1 mark).
Disadvantage for CISC (up to 1 mark).

<u>Indicative content</u>
Key Differences:

RISC Architecture (Up to 1 mark):
RISC focuses on a simplified set of instructions (1 mark), each of which is designed to execute in a single clock cycle (1 mark). This simplifies hardware design (1 mark). It typically relies on pipelines and uses more general-purpose registers (1 mark).

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CISC Architecture (Up to 1 mark):
CISC employs a larger, more complex set of instructions (1 mark) which may take multiple clock cycles (1 mark). This reduces software complexity (1 mark).

Disadvantages (Up to 2 marks – 1 for RISC, 1 for CISC):

- ***RISC: The simplified instruction set often increases the number of instructions necessary for complex operations (1 mark), which can make code size larger (1 mark) and increase memory usage (1 mark).***
- ***CISC: The complex instruction set increases hardware design complexity (1 mark), leading to potentially higher costs (1 mark) and slower clock speeds (1 mark).***

c) Which architecture in b) would the processor of a new smartphone likely utilise? **1**

Mark Scheme:

RISC Architecture (1 mark)

Total 10 Marks

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Question 5

a) Computer systems process binary numbers which can be expressed in different formats. **3**

Add binary numbers 10101 and 00100. Give the result in base 2. Show your working.

Mark Scheme:

	16	8	4	2	1
	1	0	1	0	1
+	0	0	1	0	0
	1	**1**	**0**	**0**	**1**
		1			

1 mark for displaying workings

1 mark for evidence of carry

1 mark for correct answer

b) Convert the hexadecimal number F8D4 to its decimal value. Show your working. **3**

Mark Scheme:

F = 15, 8 =8, d =13, 4 =4 (1 mark)

= $(15 \times 16^3) + (8 \times 16^2) + (13 \times 16^1) + (4 \times 16^0)$ (1 mark)
= 63700 (1 mark for correct answer)

Accept other logical ways of converting.

c) Represent the number 1874 in 2's complement. Show your working **3**

Mark Scheme:

		1024	512	256	128	64	32	16	8	4	2	1
1874		**1**	**1**	**1**	**0**	**1**	**0**	**1**	**0**	**0**	**1**	**0**
Flip		**0**	**0**	**0**	**1**	**0**	**1**	**0**	**1**	**1**	**0**	**1**
+1		**0**	**0**	**0**	**1**	**0**	**1**	**0**	**1**	**1**	**1**	**0**
Sign	**1**	**0**	**0**	**0**	**1**	**0**	**1**	**0**	**1**	**1**	**1**	**0**

1 mark for flip of bits

1 mark for adding 1 to binary

1 mark for correct sign applied.

d) Which number system is also known as the base-10 system? **1**

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Mark Scheme:

The decimal number system is also known as the base-10 system (1 mark)

Total 10 Marks

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Question 6

a) Explain the key differences between compilers and interpreters in terms of how they can help execute programs. For each, highlight how these differences affect their functionality. **4**

Mark scheme:

Compiler (Up to 2 Marks – 1 for execution, 1 for functionality)

A compiler translates the entire source code of a program into machine code (executable) upfront, before the program is executed. (1 mark)

Because compilers perform complete translation upfront, they result in faster program execution after the initial compilation is complete (1 mark).

Interpreter (Up to 2 Marks – 1 for execution, 1 for functionality)

An interpreter translates and executes source code line-by-line or statement-by-statement during runtime (1 mark).

Interpreters execute code immediately, which allows for quick debugging and testing since errors are reported as encountered (1 mark).

b) What is system software? Give TWO (2) key differences between system software and application software. **4**

In your answer, you should focus on each software's *purpose* and *dependency*.

Mark Scheme:

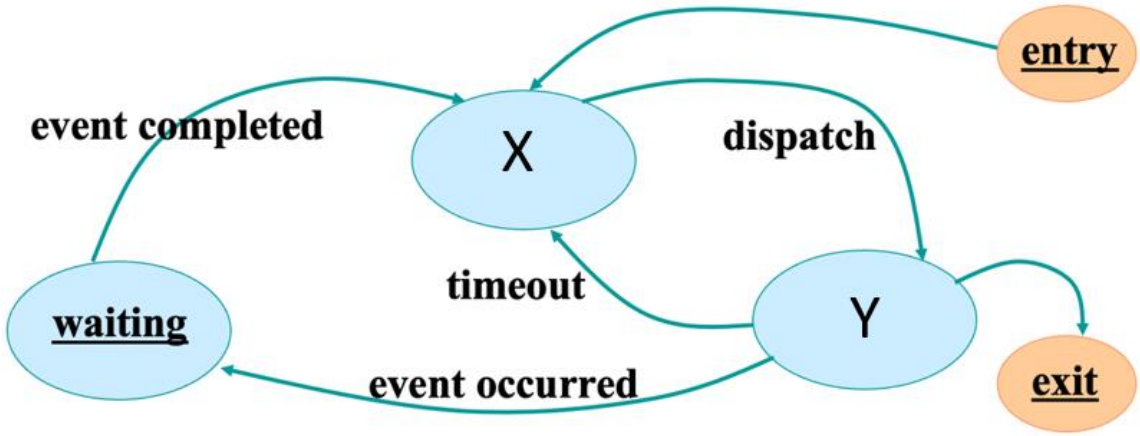
System Software:
Purpose: Manages system resources and provides a platform for applications. (1 mark)
Dependency: Runs independently and is essential for the system. (1 mark)

Application Software:
Purpose: Designed for end-users to perform specific tasks. (1 mark)
Dependency: Depends on system software to function. (1 mark)

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c) Processes in a computer system can be in ONE(1) of several states. **2**

In the following diagram, state the missing states (denoted by X and Y).



Mark Scheme:

X – Ready (1 mark)
Y – Running (1 mark)

Total 10 Marks

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Question 7

a) Name the THREE (3) main stages of a CPU's instruction cycle. **3**

Mark Scheme:

Fetch instruction from RAM (1 mark)
Decode instruction (1 mark)
Execute instruction (1 mark)

b) Explain the purpose of an opcode. **2**

Mark Scheme:

An opcode (operation code) is part of an instruction in a CPU's instruction set that specifies the operation to be performed. (1 mark)

The opcode tells the CPU which operation to execute, such as addition, subtraction, data transfer, or logical operations. (1 mark)

c) Explain the purpose of an operand in a CPU. Give ONE (1) example of an operation involving an operand. **3**

Mark Scheme:

Definition of operand (1 mark) – An operand in a CPU is a value or memory location that an instruction operates on during execution.

Explanation of its role (1 mark) – The operand provides the necessary data for the CPU to perform computations or logical operations.

Example of an operation (1 mark) – Typically mathematical operations such as addition, subtraction, division OR multiplication.

d) Before the CPU processes any data, it is checked via a parity checker. **2**

Describe how a parity checker works.

Mark Scheme:

A parity checker for a CPU is a component or circuitry responsible for verifying the integrity of data transmitted to or stored in the CPU's memory (1 mark). It works by adding an extra parity bit to each byte of data, which is calculated based on the number of bits set to 1 in the byte (either even or odd parity). (1 mark)

Total 10 Marks

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Question 8

a) Is the following statement TRUE or FALSE? **1**

In a multi-core computer system, each core can execute independent tasks simultaneously.

Mark scheme:
True (1 mark)

b) A CPU, with its pipeline switched off, executes 2500 instructions in 5000 clock cycles. What is the average clock cycle per instruction executed? **1**

Mark scheme:
5000 / 2500 = 2 instructions per clock cycle (1 mark)

c) Outline FIVE (5) distinct characteristics of distributed computing. **5**

Mark scheme:

Any 5 from the following:
Multiple computer systems working together to solve complex problems (1 mark)
Combining multiple computer systems computational power (1 mark)
Resource Sharing (both hardware and software) (1 mark)
Remote process management (1 mark)
Increased Reliability (or Fault Tolerance) (1 mark)
Scalability (or Incremental Growth) (1 mark)
Multiple platform interoperability (1 mark)

Answers may differ from the above suggestions. There may be alternative suitable answers provided by students.

d) Micro-controllers are widely used in a number of different areas. Provide THREE (3) examples of application areas where micro-controllers are used. **3**

Mark Scheme:

Any 3 from below:
- ***Automotive industry (1 mark)***
- ***Security industry (1 mark)***
- ***Domestic appliances (1 mark)***
- ***Military equipment (1 mark)***
- ***Scientific instruments (1 mark)***
- ***Computer peripherals (1 mark)***
- ***Entertainment industry (1 mark)***

Accept any other reasonable answers.

Total 10 Marks

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Question 9

a) Explain what is meant by the term network topology. Provide TWO (2) examples of different network topologies and state their *fault tolerance* and *performance* characteristics. **7**

Mark scheme

A network topology refers to the physical or logical arrangement of devices and cables in a network. (1 mark)

Any 2 from:

- ***BUS Topology (1 mark)***

Fault Tolerance: Failure in the backbone cable affects the entire network. (1 mark)
Performance: Performance decreases with more devices due to collisions. (1 mark)

- ***TREE Topology (1 mark)***

Fault Tolerance: Faults in a sub-branch won't affect the whole network, but a root failure disrupts everything. (1 mark)
Performance: Hierarchical structure supports segmentation but depends on root node efficiency. (1 mark)

- ***STAR Topology (1 mark)***

Fault Tolerance: Individual device failures don't affect the network, but a hub failure disrupts communication. (1 mark)
Performance: Dedicated connections ensure fast data transmission and minimize collisions. (1 mark)

- ***MESH Topology (1 mark)***

Fault Tolerance: Redundant paths ensure the network remains operational even if multiple links fail. (1 mark)
Performance: Data travels quickly due to multiple paths, but greater complexity can slow scalability. (1 mark)

- ***HYBRID Topology (1 mark)***

Fault Tolerance: Combines strengths of constituent topologies but inherits some weaknesses of individual systems. (1 mark)
Performance: Performance depends on how the topologies are combined and utilised. (1 mark)

Answers may differ from the above, but if the fundamental points are valid accept variants.

b) State how a router differs from a switch in terms of functionality. **2**

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Mark scheme

A router connects different networks (e.g., LAN to the internet). (1 mark)
A switch operates within a single network (LAN). (1 mark)

Accept any other reasonable answers.

c) Name the widely adopted communication protocol responsible for allowing emails to be sent across the internet. **1**

Mark scheme

POP, POP3, IMAP, or SMTP (1 mark)

Total 10 Marks

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Question 10

a) What is the octal equivalent of the hexadecimal value B35? Show your working. **3**

Mark Scheme:

Hex to Binary
B = 1011, 3 = 0011, 5 = 0101
B35 (hex) = 1011 0011 0101
(1 mark for Hex to Binary)

Binary to Octal
101 – 100 – 110 – 101
101 = 5,
100 = 4,
110 = 6,
101 = 5
(1 mark Binary to Octal)
Combine the octal digits: 5465_8 (1 mark for correct answer)

Accept other logical ways of converting.

b) Complete the following table for the logic function: A.(B+C) **7**

A	***B***	***C***	***B + C***	**A . (B + C)**
0	*0*	*0*		
0	*0*	*1*		
0	*1*	*0*		
0	*1*	*1*		
1	*0*	*0*		
1	*0*	*1*		
1	*1*	*0*		
1	*1*	*1*	1	1

Mark scheme

1 mark per correct row (indicated within bold borders) = 7 marks

A	***B***	***C***	***B OR C***	***A AND (B OR C)***
0	*0*	*0*	***0***	***0***
0	*0*	*1*	***1***	***0***
0	*1*	*0*	***1***	***0***
0	*1*	*1*	***1***	***0***
1	*0*	*0*	***0***	***0***
1	*0*	*1*	***1***	***1***
1	*1*	*0*	***1***	***1***
1	*1*	*1*	*1*	*1*

Total 10 Marks

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End of paper

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Learning Outcomes Matrix

Question	Learning Outcomes assessed	Marker can differentiate between varying levels of achievement
1	a-LO1, b-LO7, c-LO7, d-LO6, e-LO1, f-LO3, g-LO3, h-LO4, i-LO7, j-LO5	Yes
2	a-LO1, b-LO1	Yes
3	a-LO2,5	Yes
4	a-LO6, b/c-LO4	Yes
5	a/b/c-LO3	Yes
6	a-LO3, b-LO5, c-LO2	Yes
7	a/b/c/d-LO3	Yes
8	a-LO6, b-LO4, c-LO6, d-LO6	Yes
9	a/b/c-LO-7	Yes
10	a/b-LO3	Yes

Grade descriptors

Learning Outcome	Fail	Referral	Pass	Merit	Distinction
Understand and identify the main types and components of computer systems	None or inadequate description of computer systems and components	Some credible but incomplete attempt to demonstrate understanding of computer systems and components	Limited but satisfactory demonstration of understanding of computer systems and components	Good demonstration of understanding of computer systems and components	Excellent demonstration of understanding of computer systems and components
Describe the structure and role of modern operating systems	None or very basic description of components of operating systems	A very limited and basic description of components of operating systems	Limited and above basic description of components of operating systems	Good description of most components of operating systems	Excellent and full description of all components of operating systems
Understand and work with binary numbers and computer logic	No demonstration of working with binary numbers and logic operators	A basic demonstration of restricted range of working with binary numbers and logic operators	An above basic demonstration of working with restricted range of binary numbers and logic operators	A good demonstration of working with a wide range of binary numbers and logic operators	An excellent demonstration of working with full range of binary numbers and logic operators
Understand the essential structure of computer programs	None or trivial explanation of instruction sets, addressing modes and interrupts	Limited understanding and explanation of instruction sets, addressing modes and interrupts	Basic understanding and explanation of instruction sets, addressing modes and interrupts	Good understanding and explanation of instruction sets, addressing modes and interrupts	Excellent understanding and explanation of instruction sets, addressing modes and interrupts

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Explain the hierarchy of computer software	No or wrong explanation of computer software hierarchy	Limited and very basic explanation of computer software hierarchy	Basic but limited explanation of computer software hierarchy	Good and competent explanation of computer software hierarchy	Excellent and complete explanation of computer software hierarchy
Describe alternative computer systems	No or insufficient description of performance enhancing technologies	Limited and very basic description of performance enhancing technologies	Basic and partly satisfactory description of performance enhancing technologies	Good description of performance enhancing technologies	Excellent and complete description of performance enhancing technologies
Discuss the role of Data Communications and Networks in computer systems	No or very trivial discussion of data communications protocols and components	A very basic discussion of data communications protocols and components	Basic but credible discussion of data communications protocols and components	Good and wide-ranging discussion of data communications protocols and components	Excellent and complete discussion of data communications protocols and components

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